Foundations of Deep Learning

Lecture 00

Prerequisites

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Prerequisites: Table of Content

Section 1

VECTORS, MATRICES AND NOTATIONS

Vectors

A **column vector** is a $d \times 1$ array that is, an array consisting of a single column of d elements, denoted as

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} .$$

Similarly, a **row vector** is a $1 \times d$ array that is, an array consisting of a single row of d elements, denoted as

$$\mathbf{x} = \begin{bmatrix} x_1 & x_2 & \dots & x_d \end{bmatrix} .$$

The transpose (indicated by $\top$) of a row vector is a column vector.

Vectors

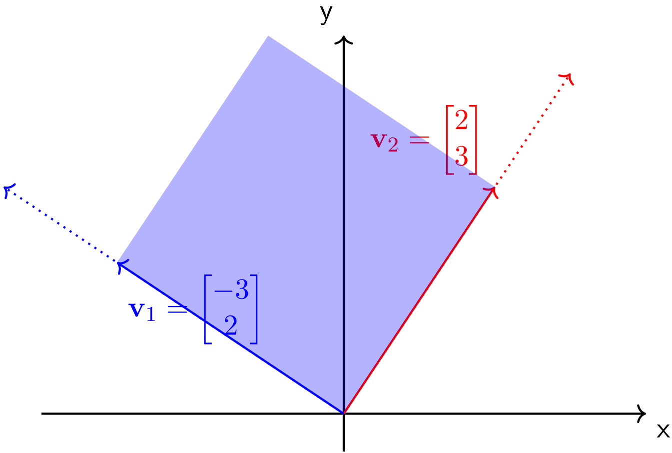
Inner product The inner product between two vectors $\mathbf{u} \in \mathbb{R}^d$ and $\mathbf{v} \in \mathbb{R}^d$ is defined as

$$\mathbf{u}^\top \mathbf{v} = \sum_{i=1}^{d} u_i v_i.$$

Span The **span** of a set of vectors is the set of all possible linear combinations of those vectors. If we have a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ in a **vector space** V, the span of these vectors, denoted by $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$, is defined as:

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k) = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k \mid c_1, c_2, \ldots, c_k \in \mathbb{R}\}.$$

Span: example



$$\underbrace{\begin{pmatrix} -3 & 2 \\ 2 & 3 \end{pmatrix}}_{\mathbf{V}} \begin{pmatrix} 0.5 \\ 1 \end{pmatrix} = 0.5 \underbrace{\begin{pmatrix} -3 \\ 2 \end{pmatrix}}_{\mathbf{v}_1} + 1 \underbrace{\begin{pmatrix} 2 \\ 3 \end{pmatrix}}_{\mathbf{v}_1} = \begin{pmatrix} 0.5 \\ 4 \end{pmatrix}$$

Span

Linearly dependent columns

$$\underbrace{\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}}_{\mathbf{V}} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = 1 \underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_{\mathbf{v}_1} + 0.5 \underbrace{\begin{pmatrix} 2 \\ 4 \end{pmatrix}}_{\mathbf{v}_1} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

Terminology

Definition 1
The column space (also called the range or image) of a matrix $\mathbf{A}$ is the span (set of all possible linear combinations) of its column vectors.

Matrices

Matrix = rectangular array of numbers.

Matrices will be denoted by a boldface capital letter, for instance

$$\mathbf{A} = \begin{pmatrix} A_{11} & \dots & A_{1d} \\ \vdots & \ddots & \vdots \\ A_{p1} & \dots & A_{pd} \end{pmatrix}, \quad \mathbf{A}^\top = \begin{pmatrix} A_{11} & \dots & A_{p1} \\ \vdots & \ddots & \vdots \\ A_{1d} & \dots & A_{pd} \end{pmatrix}$$

Multiplication The multiplication of two matrices $\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times d}$ is

$$\mathbf{C} = \mathbf{AB}, \quad C_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}.$$

Matrix Multiplication Properties

The matrix multiplication operation has the following properties:

- **Distributive property:** Matrix multiplication is distributive over matrix addition. For matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ of compatible dimensions,

$$\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC} \quad \text{and} \quad (\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC}+\mathbf{BC}.$$

- **Associative property:** Matrix multiplication is associative. For matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ of compatible dimensions,

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}).$$

- **Not commutative:** Matrix multiplication is generally not commutative. For matrices $\mathbf{A}$ and $\mathbf{B}$ of compatible dimensions,

$$\mathbf{AB} \neq \mathbf{BA} \quad \text{in general.}$$

Inverse of a Matrix

The **inverse** of a matrix $\mathbf{A}$ is a matrix, denoted by $\mathbf{A}^{-1}$, such that when it is multiplied by $\mathbf{A}$, it yields the identity matrix.
Specifically, for an $n \times n$ matrix $\mathbf{A}$,

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n,$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix.

Notation

- Bold small letter $\mathbf{x}$ is a column vector.
- Bold capital letter $\mathbf{A}$ is a matrix.
- $\mathbf{x}^\top, \mathbf{A}^\top$: transpose of a vector (i.e. a row vector) and a matrix respectively.
- $\mathbb{R}$: reals.
- $a := b$: a is defined by b.
- $\frac{\partial f}{\partial x}$: partial derivative of a function f with respect to x.
- $\frac{df}{dx}$: total derivative of a function f with respect to x.
- ∇: gradient.
- $\| \cdot \|$: norm (by default $\|\mathbf{x}\| = \|\mathbf{x}\|_2$).

Math Symbols

- $C(X, Y)$: Space of continuous functions $f : X \to Y$.
- $C(\mathbb{R})$: space of continuous functions $f : \mathbb{R} \to \mathbb{R}$, usually endowed with the uniform norm topology.
- $C_c(\mathbb{R})$: continuous functions with compact support.
- $C([a, b])$: space of all continuous functions that are defined on a closed interval $[a, b]$.
- $C_b(\mathbb{R})$: space of continuous bounded functions.
- $B(\mathbb{R})$: bounded functions.
- $C_0(\mathbb{R})$: continuous functions which vanish at infinity.
- $C^r(\mathbb{R})$: continuous functions that have continuous first r derivatives.
- $C^\infty(\mathbb{R})$: smooth functions.
- $C_c^\infty(\mathbb{R})$: smooth functions with compact support.

Math Symbols

- $\mathbb{F}$: Field of either real $\mathbb{R}$ or complex numbers $\mathcal{C}$.
- ℓ^p: used to indicate a p-summable discrete set of values.
  - ℓ^1: the space of sequences whose series is absolutely convergent.
  - ℓ^2: the space of square-summable sequences, which is a Hilbert space.
  - ℓ^∞: the space of bounded sequences.
- L^p: used to indicate p-summable functions (with respect to some measure) on a non-discrete measure space.
- $\mathbb{E}[X]$: The expectation of a random variable X.
- $\mathbf{A} \succcurlyeq 0$: $\mathbf{A}$ is positive semidefinite.

Section 2

Vector Spaces

Vector Space

A **vector space** (also called a linear space) is a collection of objects called **vectors**, which may be added together and multiplied by scalars.

Formally, a vector space over a field F (e.g. the field of real numbers) is a set V together with two operations (addition, multiplication) that satisfy several axioms:

- *Associativity*: For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
- *Commutativity*: For all $\mathbf{u}, \mathbf{v} \in V$, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- *Identity*: There exists an element $\mathbf{0} \in V$ such that for all $\mathbf{u} \in V$, $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
- *Inverse*: For every $\mathbf{u} \in V$, there exists an element $-\mathbf{u} \in V$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

Example: The Vector Space $\mathbb{R}^2$

Consider the set $\mathbb{R}^2$, which consists of all ordered pairs of real numbers (x, y). We define two operations on $\mathbb{R}^2$:

- **Vector Addition**:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2).$$

- **Scalar Multiplication**:

$$a \cdot (x, y) = (ax, ay),$$

  where a is a scalar (a real number).

Norm

A normed vector space is a space equipped with a norm. Informally this means that we need to be able to add vectors and scale them with a scalar.

Norm: Formal Definition

Given the definition of a vector space V, we can define a norm as a function from V to $\mathbb{R}$.

Definition 2
Given a vector space V over a subfield F of the complex numbers, a norm on V is a nonnegative-valued scalar function $p : V \to [0, +\infty)$ with the following properties:

- For all $v \in V$, $p(v) \geq 0$ (non-negativity)
- If $p(v) = 0$ then $v = 0$ (positive definite)
- $p(av) = |a|p(v)$ (absolutely homogeneous)
- $p(u + v) \leq p(u) + p(v)$ (triangle inequality)

Example 1: Vector Norm

A vector norm is a function that assigns a non-negative scalar value to a vector in a vector space, which intuitively represents the length or size of the vector. More formally, if $\mathbf{v}$ is a vector in a vector space V, a norm $\|\mathbf{v}\|$ satisfies the following properties:

i) **Non-negativity (Positivity)**:

$$\|\mathbf{v}\| \geq 0 \quad \text{and} \quad \|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}$$

ii) **Scalar Multiplication**:

$$\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\|, \quad \alpha \in \mathbb{R}.$$

iii) **Triangle Inequality**:

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|, \quad \mathbf{u}, \mathbf{v} \in V.$$

Common Examples of Vector Norms

- **Euclidean norm (or 2-norm)**:

$$\|\mathbf{v}\|_2 = \left(\sum_{i=1}^{n} |v_i|^2\right)^{1/2}$$

  where $\mathbf{v} = (v_1, v_2, \ldots, v_n)$.

- **1-norm (or Manhattan norm)**:

$$\|\mathbf{v}\|_1 = \sum_{i=1}^{n} |v_i|$$

- **Infinity norm (or maximum norm)**:

$$\|\mathbf{v}\|_\infty = \max_{1 \le i \le n} |v_i|$$

Example 2: Matrix Norm

Suppose a vector norm $\|\cdot\|$ on $\mathbb{R}^m$ is given (e.g. the Euclidean norm). Any $m \times n$ matrix $\mathbf{A}$ induces a linear operator from $\mathbb{R}^n$ to $\mathbb{R}^m$ with respect to the standard basis, and one defines the corresponding operator norm on the space $\mathbb{R}^{m \times n}$:

$$\|\mathbf{A}\|_p = \sup \left\{ \frac{\|\mathbf{A}\mathbf{x}\|_p}{\|\mathbf{x}\|_p} : \mathbf{x} \in \mathbb{R}^n \text{ with } \mathbf{x} \neq 0 \right\}.$$

In practice, one is often interested in the case $p = 2$, for which

$$\begin{aligned} \|\mathbf{A}\|_2 &= \sup \left\{ \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} : \mathbf{x} \in \mathbb{R}^n \text{ with } \mathbf{x} \neq 0 \right\} \\ &= \sigma_1(\mathbf{A}), \end{aligned}$$

where $\sigma_1(\mathbf{A})$ is the largest singular value of $\mathbf{A}$.

Cauchy-Schwarz inequality

Proposition 3

The Cauchy-Schwarz inequality states that for all vectors $\mathbf{x}$ *and* $\mathbf{y}$ *of an inner product space, the following holds:*

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$

Writing $\mathbf{x} = [x_1 \dots x_n]$ *and* $\mathbf{y} = [y_1 \dots y_n]$, *this can also be written as*

$$\left(\sum_{i=1}^{n} x_i y_i \right)^2 \leq \sum_{i=1}^{n} x_i^2 \sum_{i=1}^{n} y_i^2.$$

Equality occurs if and only if there exists a constant c *such that* $x_i = c y_i \; \forall i = 1, \dots n$.

Section 3

Functions

Continuous Function

Definition 4
A function $f(x)$ is continuous at $x = c$ if:

$$\lim_{x \to c} f(x) = f(c)$$

Example 5
The function $f(x) = x^2$ is continuous for all real numbers because for any point c:

$$\lim_{x \to c} x^2 = c^2 = f(c)$$

Gradients

Consider a real-valued (univariate) function $f : \mathbb{R} \to \mathbb{R}$. Its **derivative** is defined by

$$f'(x) := \lim_{\epsilon \to 0} \frac{f(x+\epsilon) - f(x)}{\epsilon}.$$

Let's generalize this definition to the case where the function f is multivariate, i.e. $f : \mathbb{R}^d \to \mathbb{R}$. In this case, we have one (partial) derivative for each dimension, i.e. $\frac{\partial f}{\partial x_i}$ defined as

$$\frac{\partial f}{\partial x_i} := \lim_{\epsilon \to 0} \frac{f(x_1, \ldots, x_i + \epsilon, \ldots, x_d) - f(x_1, \ldots, x_i, \ldots, x_d)}{\epsilon}.$$

The function f is **differentiable** if the limit exists.

Gradients

The **gradient** denoted by $\nabla f(\mathbf{x})$ gives us a way to pack all partial derivatives into one vector. Denoting by $\mathbf{x} = (x_1, x_2, \ldots x_d)$, we then define the gradient as

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{pmatrix}$$

where we can also define

$$\frac{\partial f}{\partial x_i} := \lim_{\epsilon \to 0} \frac{f(\mathbf{x} + \epsilon \mathbf{e}_i) - f(\mathbf{x})}{\epsilon},$$

with $\mathbf{e}_i$ being a standard basis vector (composed of zeros and a single one at the i-th coordinate).

Jacobian Matrix

Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a vector-valued function,

$$f(\mathbf{x}) = \begin{pmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{pmatrix}.$$

The Jacobian matrix $J_f(\mathbf{x})$ is the $m \times n$ matrix of first-order partial derivatives:

$$J_f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}.$$

Chain Rule

The chain rule is a formula to compute the derivative of a composite function $f(g(x))$.

For functions of single variables, i.e. $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$, the chain rule is

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

Alternatively, it can also be written as

$$\frac{d}{dx} f(g(x)) = \frac{df}{dg} \cdot \frac{dg}{dx},$$

where dg is an abbreviation for $dg(x)$.

General Case of Chain Rule

Let's consider the general case where $\mathbf{f} : \mathbb{R}^m \to \mathbb{R}^d$ and $\mathbf{g} : \mathbb{R}^k \to \mathbb{R}^m$. Then the composed function is $f(g(\mathbf{x})) : \mathbb{R}^k \to \mathbb{R}^d$.

Chaine rule

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{f}(\mathbf{g}(\mathbf{x})) = \frac{\partial \mathbf{f}}{\partial \mathbf{g}}(g(\mathbf{x})) \cdot \frac{\partial \mathbf{g}}{\partial \mathbf{x}}(\mathbf{x}) \in \mathbb{R}^{d \times k},$$

where $\frac{\partial \mathbf{f}}{\partial \mathbf{g}}(g(\mathbf{x})) \in \mathbb{R}^{d \times m}$ is the Jacobian matrix of $\mathbf{f}$ w.r.t. $\mathbf{g}$ and $\frac{\partial \mathbf{g}}{\partial \mathbf{x}}(\mathbf{x}) \in \mathbb{R}^{m \times k}$ is the Jacobian matrix of $\mathbf{g}$ w.r.t. $\mathbf{x}$.

Taylor's Theorem (scalar version)

Taylor's theorem gives an approximation of a k-times differentiable function f around a given point.

Theorem 6 (Taylor's theorem)

Let $k \geq 1$ be an integer and let the function $f : \mathbb{R} \to \mathbb{R}$ be k times differentiable at the point $a \in \mathbb{R}$. Then there exists a function $h : \mathbb{R} \to \mathbb{R}$ such that

$$\begin{aligned} f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots \\ + \frac{f^{(k)}(a)}{k!}(x-a)^k + h_k(x)(x-a)^k, \end{aligned}$$

and

$$\lim_{x \to a} h_k(x) = 0.$$

Proof Idea of Taylor's Theorem

We prove it for $k = 3$ as the generalization follows easily.

Note that we assumed that the function f is k-times differentiable. We can therefore repeatedly apply the fundamental theorem of calculus as follows:

$$\begin{aligned}
f(x) &= f(a) + (f(x) - f(a)) = f(a) + \int_a^x f'(x_1)\, dx_1 \\
&= f(a) + \int_a^x f'(a) + (f'(x_1) - f'(a))\, dx_1 \\
&= f(a) + f'(a)(x-a) + \int_a^x (f'(x_1) - f'(a))\, dx_1 \\
&= f(a) + f'(a)(x-a) + \int_a^x \int_a^{x_1} f^{(2)}(x_2)\, dx_2\, dx_1 \\
&= \dots
\end{aligned}$$

Mean Value Theorem

By choosing $k = 1$ in Taylor's theorem, we recover the mean value theorem.

Theorem 7 (Mean value theorem (scalar version))

Let $f : [a, b] \to \mathbb{R}$ be a continuous function on the closed interval $[a, b]$, and differentiable on the open interval (a, b), where $a < b$. Then there exists some $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Multivariable Functions

For multivariate function $f : \mathbb{R}^d \to \mathbb{R}$, we have for some $c \in (a, b)$,

$$
\begin{aligned}
&f(\mathbf{x}+\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x}+c\mathbf{y})^\top \mathbf{y} \\
\implies &f(\mathbf{x}+\mathbf{y}) - f(\mathbf{x}) = \nabla f(\mathbf{x}+c\mathbf{y})^\top \mathbf{y},
\end{aligned}
$$

or in an integral form stated in the theorem below.

Theorem 8 (Mean value theorem (multivariable version))

Given a continuously differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ and any vector $\mathbf{y} \in \mathbb{R}^d$, we have that

$$
f(\mathbf{x}+\mathbf{y}) - f(\mathbf{x}) = \int_0^1 \nabla f(\mathbf{x}+t\mathbf{y})^\top \mathbf{y}\, dt.
$$

Proof of Mean Value Theorem

Define $g : [0,1] \to \mathbb{R}$ as $g(t) = f(\mathbf{x} + t\mathbf{y})$. Then by applying the fundamental theorem of calculus to g, we get

$$f(\mathbf{x}+\mathbf{y}) - f(\mathbf{x}) = g(1) - g(0) = \int_0^1 g'(t)\,dt.$$

Let $u(t) = \mathbf{x} + t\mathbf{y}$. By the multivariate chain rule,

$$g'(t) = \sum_{i=1}^{d} \frac{\partial f(\mathbf{x}+t\mathbf{y})}{\partial u_i} \cdot \frac{\partial u_i}{\partial t} = \sum_{i=1}^{d} \frac{\partial f(\mathbf{x}+t\mathbf{y})}{\partial u_i} \cdot y_i = \nabla f(\mathbf{x}+t\mathbf{y})^\top \mathbf{y}.$$

Therefore

$$f(\mathbf{x}+\mathbf{y}) - f(\mathbf{x}) = \int_0^1 \nabla f(\mathbf{x}+t\mathbf{y})^\top \mathbf{y}\,dt.$$

High-order Variant of Mean Value Theorem

A similar expression to the mean value theorem can be stated for twice differentiable functions:

$$f(\mathbf{x}+\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{y} + \frac{1}{2}\int_0^1 \mathbf{y}^\top \nabla^2 f(\mathbf{x}+t\mathbf{y})\mathbf{y}\, dt.$$

Gradient Note that the mean value theorem also applies to the gradient ∇f for functions f that are twice differentiable. It of course also applies to higher-order derivatives if they exist. For the first derivative, we simply get:

$$\nabla f(\mathbf{x}+\mathbf{y}) = \nabla f(\mathbf{x}) + \int_0^1 \nabla^2 f(\mathbf{x}+t\mathbf{y})^\top \mathbf{y}\, dt.$$

Calculating Gradients

Example: Gradient of a Multivariable Function

Consider the function $f(\mathbf{x}) = \mathbf{A}\mathbf{x} \in \mathbb{R}^p$, where $\mathbf{x} \in \mathbb{R}^d, \mathbf{A} \in \mathbb{R}^{p \times d}$.
Our goal is to compute the gradient $\frac{df}{d\mathbf{x}}$.
First, note that the dimension of the gradient $\frac{df}{d\mathbf{x}}$ is $\mathbb{R}^{p \times d}$. Let's compute the partial derivative of f w.r.t. a single x_j. We have

$$f_i(\mathbf{x}) = \sum_{j=1}^{d} A_{ij} x_j \implies \frac{\partial f_i}{\partial x_j} = A_{ij}.$$

Collecting all the partial derivatives in the Jacobian, we obtain the following expression for the gradient:

$$\frac{df}{d\mathbf{x}} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_p}{\partial x_1} & \cdots & \frac{\partial f_p}{\partial x_d} \end{pmatrix} = \begin{pmatrix} A_{11} & \ldots & A_{1d} \\ \vdots & \ddots & \vdots \\ A_{p1} & \ldots & A_{pd} \end{pmatrix} = \mathbf{A} \in \mathbb{R}^{p \times d}.$$

Lipschitz Property

Intuitively, a Lipschitz continuous function is limited in how fast it can change.

Definition 9

The function $f : \mathbb{R}^d \to \mathbb{R}$ is L-Lipschitz continuous if:

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq L \|\mathbf{x} - \mathbf{y}\| \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d.$$

Lipschitz Property

Lemma 10

A function $f(\mathbf{x})$ is L-Lipschitz continuous if its gradient is bounded by L.

Proof.

By the mean value theorem, we have :

$$f(\mathbf{x}) - f(\mathbf{y}) = \nabla f(\mathbf{y} + c(\mathbf{x} - \mathbf{y}))^\top (\mathbf{x} - \mathbf{y})$$

Therefore,

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq \sup_{\mathbf{z} \in \mathbb{R}^d} \|\nabla f(\mathbf{z})\| \|\mathbf{x} - \mathbf{y}\| \leq L \|\mathbf{x} - \mathbf{y}\|.$$

$\square$

Smoothness

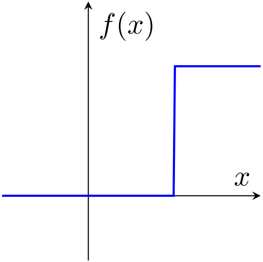
We say that a differentiable function is smooth if its gradient is Lipschitz continuous. We formalize this below.

Definition 11

The function $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable with L-Lipschitz-continuous gradient, i.e.

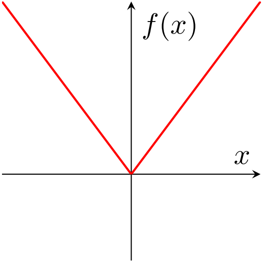
$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\| \, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d.$$

Examples of Continuous and Differentiable Functions



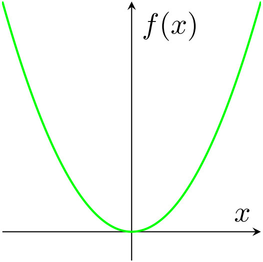
(a) $f(x) = \begin{cases} 0 & x < 1 \\ 1 & x \geq 1 \end{cases}$

Continuous: No
Differentiable: No



(b) $f(x) = |x|$

Continuous: Yes
Differentiable: No



(c) $f(x) = x^2$

Continuous: Yes
Differentiable: Yes

Section 4

Linear Algebra

Eigenvalues

Consider a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ (recall that eigenvalues are not defined for rectangular matrices, for that we need the concept of singular values).

Definition $\lambda \in \mathbb{R}$ is an eigenvalue of $\mathbf{A}$ and $\mathbf{v} \in \mathbb{R}^n \backslash \{\mathbf{0}\}$ is an eigenvector of $\mathbf{A}$ if the following holds:

$$\mathbf{A}\mathbf{v} = \lambda \mathbf{v}.$$

Note that this relation is also equivalent to $(\mathbf{A} - \lambda \mathbf{I})\mathbf{v} = 0$ or $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$.

Convention Assume the λ_i's are sorted as follows:

$$\lambda_1(\mathbf{A}) \geq \ldots \geq \lambda_n(\mathbf{A}).$$

Trace of a matrix

The trace of a square matrix $\mathbf{A}$, denoted as $\mathrm{tr}(\mathbf{A})$, is the sum of its diagonal elements. If $\mathbf{A}$ is an $n \times n$ matrix given by:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

then the trace of $\mathbf{A}$ is defined as:

$$\mathrm{tr}(\mathbf{A}) = \sum_{i=1}^{n} a_{ii}.$$

The trace of a matrix is also equal to the sum of its eigenvalues. If $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of $\mathbf{A}$, then:

$$\mathrm{tr}(\mathbf{A}) = \sum_{i=1}^{n} \lambda_i.$$

Properties of the Trace

- **Linearity**: For any two $n \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$, and any scalars α and β,

$$\operatorname{tr}(\alpha\mathbf{A} + \beta\mathbf{B}) = \alpha \operatorname{tr}(\mathbf{A}) + \beta \operatorname{tr}(\mathbf{B})$$

- **Cyclic property**: For any $n \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$,

$$\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$$

- **Trace of a transpose**: For any $n \times n$ matrix $\mathbf{A}$,

$$\operatorname{tr}(\mathbf{A}^\top) = \operatorname{tr}(\mathbf{A})$$

Loewner order

PSD matrix

- A matrix $\mathbf{A}$ is positive semi-definite (PSD) if $\mathbf{u}^\top \mathbf{A}\mathbf{u} \geq 0$ for all $\mathbf{u} \in \mathbb{R}^n$,
- or equivalently all the eigenvalues λ_i of $\mathbf{A}$ are non-negative, i.e. $\lambda_i(\mathbf{A}) \geq 0 \;\; \forall\, i = 1, \ldots n$.

Loewner notation We will use the notation $\mathbf{A} \succcurlyeq 0$ to denote that $\mathbf{A}$ is PSD.

Definition 12
Let $\mathbf{A}$ and $\mathbf{B}$ be two symmetric matrices. We say that $\mathbf{A} \succcurlyeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B}$ is positive semi-definite.

Singular values

The main object of interest will now be a rectangular $m \times n$ matrix $\mathbf{A}$ with real entries. We denote by $\sigma_i(\mathbf{A})$ the i-th singular value of $\mathbf{A}$ which is equal to

$$\sigma_i(\mathbf{A}) = \sqrt{\lambda_i(\mathbf{A}\mathbf{A}^\top)} = \sqrt{\lambda_i(\mathbf{A}^\top\mathbf{A})}.$$

Remarks

- The number of nonzero singular values of $\mathbf{A}$ equals to $\operatorname{rank}(\mathbf{A}) \leq \min(m, n)$.
- Given a matrix $\mathbf{A}$, the largest singular value $\sigma_1(\mathbf{A})$ is equal to the operator norm of $\mathbf{A}$.

Singular Value Decomposition (SVD)

Theorem 13 (SVD)

Every real (or complex) matrix $\mathbf{A}$ *can be decomposed into*

$$\underset{m \times n}{\mathbf{A}} = \underset{m \times m}{\mathbf{U}} \cdot \underset{m \times n}{\mathbf{D}} \cdot \underset{n \times n}{\mathbf{V}^\top}$$

where $\mathbf{U}$, $\mathbf{V}$ *orthogonal (or unitary),* $\mathbf{D}$ *diagonal. More precisely:*

- $\mathbf{U}$ *is an* m *by* m *orthogonal matrix,* $\mathbf{U}^\top\mathbf{U} = \mathbf{U}\mathbf{U}^\top = \mathbf{I}$,
- $\mathbf{D}$ *is an* m *by* n *diagonal matrix, padded with* $\max\{m, n\} - \min\{m, n\}$ *zero rows or columns,*
- $\mathbf{V}$ *is an* n *by* n *orthogonal matrix,* $\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{I}$.

Orthogonal Vectors and Matrices

Definition 14 (Orthogonal vectors)

A set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_K\}$ in an inner product space are orthogonal if all pairwise inner products are zero, i.e.

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0, \quad (\forall i \neq j).$$

Definition 15 (Orthogonal matrix)

An orthogonal matrix $\mathbf{A}$ is a square matrix with real entries whose columns and rows are orthogonal unit vectors
(i.e. orthonormal vectors):

$$\langle \mathbf{a}_i, \mathbf{a}_j \rangle = \delta_{ij} \quad (\forall i, j).$$

Equivalently, $\mathbf{A}\mathbf{A}^\top = \mathbf{A}^\top\mathbf{A} = \mathbf{I}$.

Inverse of an Orthogonal Matrix

Theorem 16

The inverse of an orthogonal matrix is its transpose.

Proof.

It follows directly from the above equations as

$$\mathbf{A}^\top \mathbf{A} = \mathbf{I} \quad \Longrightarrow \quad \mathbf{A}^\top = \mathbf{A}^{-1}\,.$$

$\square$

SVD - Singular Values

The elements on the diagonal of $\mathbf{D}$ are called singular values and will be denoted by $\sigma_i := d_{ii}$ $(i = 1, \ldots, \min\{m, n\})$, i.e.

$$\mathbf{D} = \text{diag}\left(\sigma_1, \ldots, \sigma_{\min\{m,n\}}\right).$$

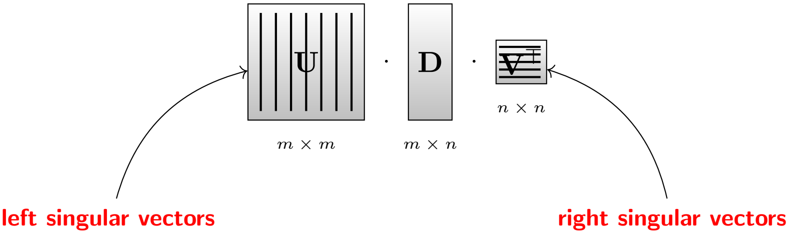
By convention, we typically order the singular values in decreasing order:

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_{\min\{m,n\}} \geq 0$$

Also note the non-negativity of singular values.

SVD - Singular Vectors

The columns of $\mathbf{U}$ are called the left singular vectors and they form an orthonormal basis for columns space of $\mathbf{A}$. Similarly, the rows of $\mathbf{V}^\top$ = columns of $\mathbf{V}$ are called the right singular vectors and form an orthonormal basis for the row space of $\mathbf{A}$.



Singular Values and Vectors

The number of distinct singular values is less than or equal to $\min\{n, m\}$. The SVD decomposition is only unique up to rotation and degeneracies (i.e. σ_i with two or more linearly independent left (or right) singular vectors).

Note that by multiplying the SVD equation by $\mathbf{V}$ one gets

$$\mathbf{AV} = \mathbf{UD} \quad \Longleftrightarrow \quad \mathbf{A}\mathbf{v}_i = \sigma_i \mathbf{u}_i \quad (\forall i \leq \min\{m, n\})$$

Similarly

$$\mathbf{A}^\top \mathbf{U} = \mathbf{V}\mathbf{D}^\top \Longleftrightarrow \mathbf{A}^\top \mathbf{u}_i = \sigma_i \mathbf{v}_i \quad (\forall i \leq \min\{m, n\})$$

Note that for the special case of $m = n$ and $\mathbf{U} = \mathbf{V}$ (i.e. $\mathbf{A}$ symmetric) we retrieve the eigendecomposition. In this case, the vectors $\mathbf{u}_i = \mathbf{v}_i$ are just the eigenvectors.

Interpretation SVD

The SVD gives us a way to decompose a linear map as **three subsequent operations**: rotation, axis scaling, and another rotation.

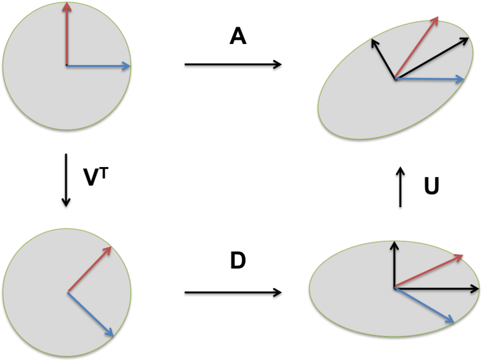


Figure: Decomposition of a linear map.

Exercise

Remark 1

If $\mathbf{A}$ *is a symmetric matrix then the singular values of* $\mathbf{A}$ *are the absolute values of the eigenvalues* λ_i *of* $\mathbf{A}$*:* $\sigma_i(\mathbf{A}) = |\lambda_i(\mathbf{A})|$.

Section 5

Probability Theory

Probability Space

Definition 17 (Probability space)

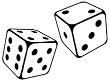
A probability space W is a unique triple $W = \{\Omega, \mathcal{F}, \Pr\}$, where Ω is its sample space, $\mathcal{F}$ its σ-algebra of events, and $\Pr$ its probability measure.

Sample Space

Definition The sample space Ω is the set of all possible samples or elementary events ω.

Example: die

Throw a die once and define the "random variable" X as "the score shown on the top face" $\rightsquigarrow$ sample space is $\Omega := \{1, 2, 3, 4, 5, 6\}$.



Other examples of sample spaces:

- Toss of a coin: $\Omega = \{H, T\}$
- Two tosses of a coin: $\Omega = \{HH, HT, TH, TT\}$
- The positive integers: $\Omega = \{1, 2, 3, \dots\}$

Sigma Algebra (σ-algebra)

The σ-algebra $\mathcal{F}$ is the set of all the **considered** events A, i.e., subsets of Ω: $\mathcal{F} = \{A | A \subseteq \Omega\}$. It also has to satisfy the following properties:

1. $\Omega \in \mathcal{F}$ and $\emptyset \in \mathcal{F}$
2. $A \in \mathcal{F} \Rightarrow \Omega \backslash A \in \mathcal{F}$ (closed under complementation)
3. $A_1, A_2, \cdots \in \mathcal{F} \Rightarrow \cup_n A_n \in \mathcal{F}$ (closed under countable unions)

Examples of Sigma Algebra (σ-algebra)

1. **The Trivial σ-Algebra:**
The smallest σ-algebra on any set X is the trivial σ-algebra, which contains only the empty set and the set itself.

$$\mathcal{A} = \{\emptyset, X\}$$

2. **The Power Set σ-Algebra:**
The largest σ-algebra on a set X is the power set of X, which contains all subsets of X.

$$\mathcal{A} = \mathcal{P}(X)$$

Examples of Sigma Algebra (σ-algebra)

3. **The Borel σ-Algebra on $\mathbb{R}$:**
The Borel σ-algebra on the real numbers $\mathbb{R}$, denoted by $\mathcal{B}(\mathbb{R})$, is generated by all open intervals $(a, b) \subset \mathbb{R}$. This σ-algebra contains all Borel sets, which are the sets that can be formed from open intervals using countable unions, countable intersections, and relative complements.

4. **σ-Algebra on a Finite Set:**
If $X = \{a, b, c\}$, one possible σ-algebra on X could be:

$$\mathcal{A} = \{\emptyset, \{a\}, \{b, c\}, \{a, b, c\}\}$$

Probability Measure

A function $\Pr : \mathcal{F} \to [0, 1]$ is called a probability measure if it satisfies the following three properties:

1. **Non-negativity:** For every $A \in \mathcal{F}$, $\Pr(A) \geq 0$.
2. **Normalization:** $\Pr(\Omega) = 1$, where Ω is the entire sample space.
3. σ**-additivity:** For any countable collection $\{A_n\}$ of pairwise disjoint sets in $\mathcal{F}$ (i.e., $A_i \cap A_j = \varnothing$ for $i \neq j$):

$$\Pr\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \Pr(A_n).$$

Probability Measure

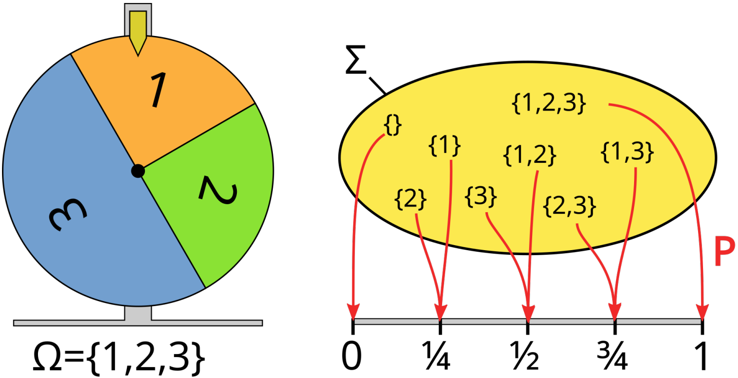


Figure: A probability measure mapping the σ-algebra for 2^3 events to the unit interval. Source: Wikipedia.

Random Variable

Informal definition A random variable is similar to a variable in mathematics, but instead of taking on just one value, it can take on a range of possible values, each with a certain probability.

Formal definition

Definition 18 (Random variable)

Given a probability space $(\Omega, \mathcal{F}, \mathrm{Pr})$, a random variable is a function from Ω to a measurable space E [a].

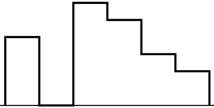
[a] A measurable space is a set equipped with a collection of subsets that are designated as measurable.

Random Variable Examples

a) Throw two dice and take $\Omega = \{1, 2, 3, 4, 5, 6\}^2$ and $\mathcal{F} = P(\Omega)$ where P is the power set, i.e. the set of all subsets of Ω (thus $\mathcal{F} = \{(1, 1), (1, 2), \ldots\}$). Then $X : \Omega \to \mathbb{R}$ defined as $(\omega_1, \omega_2) \to \omega_1 + \omega_2$ (i.e. sum the numbers on each die) is indeed a random variable.

b) Throw two dice and take $\Omega = \{1, 2, 3, 4, 5, 6\}^2$ and $\mathcal{F} = \{\emptyset, \Omega\}$. Then X as defined in a) is not a random variable. For instance $X^{-1}(\{2\}) = \{(1, 1)\} \notin \mathcal{F}$.

Discrete vs Continuous Probability Space

Discrete	Continuous
	
Probability Mass Function	Probability Density Function
$P(X = x) = f(x)$	$P(a \leq X \leq b) \leq \int_a^b f(x)\,dx$

Discrete Probability Space

Characteristic	Discrete Probability Space
Definition	Consists of a sample space Ω, a σ-algebra $\mathcal{F}$, and a probability mass function (PMF) $\Pr(X = x)$
Sample Space	Countable and finite or countably infinite, e.g. $\Omega = \{1, 2, 3, 4, 5, 6\}$ (six-sided die)
σ-algebra	Typically the power set $\mathcal{F} = P(\Omega)$
Probability Function	PMF $\Pr(X = x)$ for each x in Ω
Probability Assignment	$\Pr(X = x) \geq 0$ for all x and $\sum_{x \in \Omega} \Pr(X = x) = 1$
Probability at a Point	$\Pr(X = x) > 0$ for some x
Normalization	$\sum_{x \in \Omega} \Pr(X = x) = 1$
Random Variable	X takes values in Ω with probabilities assigned by the PMF
Example	Throwing a six-sided die: $\Pr(X = 1) = \frac{1}{6}, \Pr(X = 2) = \frac{1}{6}, \ldots, \Pr(X = 6) = \frac{1}{6}$

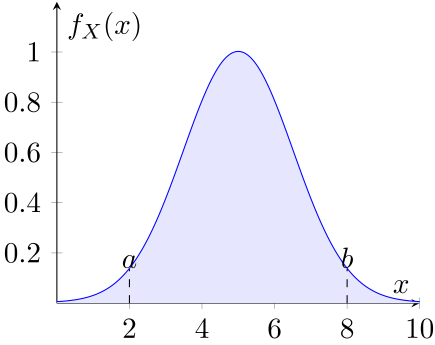
Continuous Probability Space

Characteristic	**Continuous Probability Space**
Definition	Consists of a sample space Ω, a σ-algebra $\mathcal{F}$, and a probability density function (PDF) $f(x)$
Sample Space	Uncountably infinite, e.g. $\Omega = [0, 1]$
σ-algebra	Typically the Borel set formed by Ω
Probability Function	PDF $f(x)$ such that $\Pr(a \leq X \leq b) = \int_a^b f(x)\,dx$
Probability Assignment	$f(x) \geq 0$ for all x and $\int_{-\infty}^{\infty} f(x)\,dx = 1$
Probability at a Point	$\Pr(X = x) = 0$ for all x
Normalization	$\int_{-\infty}^{\infty} f(x)\,dx = 1$
Random Variable	X takes values in Ω with probabilities assigned by the PDF
Example	Choosing a point in $[0, 1]$: $f(x) = 1$ for $0 \leq x \leq 1$, and $f(x) = 0$ elsewhere

Continuous case: PDF

A probability density function is most commonly associated with absolutely continuous univariate distributions. A random variable X has density f_X, where f_X is a non-negative Lebesgue-integrable function, if:

$$\Pr[a \leq X \leq b] = \int_a^b f_X(x)\,dx.$$



Continuous case: PDF (cont'd)

If F_X is the cumulative distribution function of X, then:

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\, du, \tag{1}$$

and (if f_X is continuous at x)

$$f_X(x) = \frac{d}{dx} F_X(x). \tag{2}$$

Intuitively, one can think of $f_X(x)\, dx$ as being the probability of X falling within the infinitesimal interval$[x, x + dx]$.

Discrete case: PMF

Similar properties hold for discrete probability spaces where the density function $f_X(x)$ is replaced by a discrete function $p : \mathcal{F} \to [0, 1]$ defined by $p_X(x) = \Pr(X = x)$.

Since the function p is defined over a discrete set, the integral is replaced by a sum, therefore for a given set A, we have

$$\Pr(A) = \sum_{\omega \in A} p_X(\omega).$$

Expectation

Continuous case If a random variable X has a continuous density $f_X(x)$, then its expectation is given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

Discrete case For a discrete random variable with a finite list $x_1, \ldots, x_k$ of possible outcomes each of which (respectively) has probability $p_1, \ldots, p_k$ of occurring, the expectation is

$$\mathbb{E}[X] = \sum_{i=1}^{k} x_i\, p_i.$$